



Project Green

K ORCHIDS · Database & System Documentation

Data model · User roles · RBAC · Workflows · Edge cases

PostgreSQL 18 · 38 tables · Express modular API · Next.js 14 web

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Companion to the Frontend Inventory

1 · System Overview

Project Green (brand **K ORCHIDS**) is a B2B wholesale platform for orchids, fertilizers and horticultural supplies. It is a monorepo with two apps:

- **apps/web** — Next.js 14 (App Router), Tailwind CSS, React Query, Axios. Six route groups, one per audience: `(public)` , `(buyer)` , `(admin)` , `(finance)` , `(inventory)` , `(delivery)` .
- **apps/api** — Express, modular by domain (auth, products, orders, rfq, invoices, payments, rma, delivery, inventory, pricing, buyers, cms, reports, settings, users, notifications, bloom). Each module is controller / service / repository / schema (Zod) / routes / tests.
- **PostgreSQL 18** — 38 tables across 8 SQL migrations. UUID PKs for identity tables, integer PKs for transactional tables.

Auth is JWT-based: a short-lived **access token** (15 min, in memory) plus a hashed **refresh token** stored as an httpOnly cookie and an `auth_sessions` row (7-day expiry, rotated on each refresh). RBAC is enforced server-side by permission code; the Next.js middleware does only coarse cookie-presence routing.

22	539	957	643	264	30	20	10
USERS	PRODUCTS	ORDERS	INVOICES	DELIVERIES	RMA	SUPPLIERS	TRADE ACCTS

Live row counts from the seeded development database.

2 · User Roles

Five fixed roles in `roles`, referenced by `users.role_id`. A user has exactly one role. Buyers additionally own a `trade_accounts` record carrying business/credit data.

Role	Purpose	Seeded	Portal
ADMIN Administrator	Full system access	1	<code>/admin</code>
TRADE_BUYER Trade Buyer	Registered wholesale buyer (self-registers, approved by admin)	10	<code>/buyer</code>
INVENTORY_MANAGER Inventory Manager	Product catalogue & stock management	1	<code>/inventory</code>
FINANCE_OFFICER Finance Officer	Invoicing, payments & financial reporting	1	<code>/finance</code>
DELIVERY_COORDINATOR Delivery Coordinator	Order dispatch & delivery tracking	1	<code>/delivery</code>

Demo credentials. Staff (admin / inventory / finance / delivery): `Staff@1234`. Buyers (`buyer@example.invalid` ... `buyer8`): `Buyer@1234`. Emails use the reserved `.invalid` TLD.

3 · RBAC — Permission Model

Authorisation is by **permission code**, not by role name. `role_permissions` maps roles → 43 permissions; `requirePermission(...codes)` middleware checks the JWT's permission list. `.own` vs `.all` codes distinguish self-scoped from global access (e.g. `order.view.own` vs `order.view.all`).

Role	# perms	Granted permissions
ADMIN	43	Full access — all 43 permissions (superuser).
TRADE_BUYER	12	product.view, stock.view, rfq.create, rfq.view.own, order.create, order.view.own, invoice.view.own, invoice.download, payment.view, rma.create, rma.view.own, delivery.view
INVENTORY_MANAGER	15	product.view/create/edit/archive, stock.view/adjust, price.change/approve, alert.view/ack, order.view.all, rfq.view.all, delivery.view, report.view, settings.view
FINANCE_OFFICER	13	invoice.view.all/generate/download, payment.view/record/reverse, credit.view, order.view.all, rfq.view.all, rma.view.all/decide, report.view, settings.view
DELIVERY_COORDINATOR	8	delivery.view/assign/update, pod.upload, order.view.all, stock.view, alert.view, settings.view

Permission catalogue (43 codes)

alert.ack · alert.view · audit.view · cms.edit · credit.view · delivery.assign · delivery.update · delivery.view · invoice.download · invoice.generate · invoice.view.all · invoice.view.own · notification.send · order.approve · order.cancel · order.create · order.view.all · order.view.own · payment.record · payment.reverse · payment.view · pod.upload · price.approve · price.change · product.archive · product.create · product.edit · product.view · report.view · rfq.create · rfq.quote · rfq.view.all · rfq.view.own · rma.create · rma.decide · rma.view.all · rma.view.own · role.assign · settings.edit · settings.view · stock.adjust · stock.view · user.manage · user.view

Time-boxed access. `role_access_windows` can restrict a role to specific weekday/time ranges (day_of_week 0–6, start_time < end_time) — enabling, e.g., "delivery coordinators may only act during depot hours."

4 · Account Lifecycle & Auth Edge Cases

Buyer onboarding

Self-register → email_verify token (24h) → email verified → trade_account = PENDING_APPROVAL → admin approves/rejects → ACTIVE (can trade) | rejected | later SUSPENDED/CLOSED

Until approved, a buyer is bounced to `/buyer/pending-approval`. The login response maps internal status to UI status: `ACTIVE→APPROVED`, `PENDING_APPROVAL→AWAITING_APPROVAL`.

Security controls (from auth.service)

- **Brute-force logout:** 5 failed attempts in 15 min → `429 ACCOUNT_LOCKED` for 15 min. Every attempt is written to `login_history` with a failure reason.
- **Session cap:** max 3 concurrent sessions per user; a 4th login evicts the oldest (configurable via `settings.sessionCap`).
- **Refresh-token rotation:** each refresh revokes the old session and issues a new token in one transaction; tokens are stored only as SHA-256 hashes.
- **Password reset:** token valid 1h; on reset **all** sessions are revoked. Changing password also revokes all sessions.
- **Email enumeration protection:** `forgot-password` always returns the same generic message whether or not the email exists.
- **Status gate:** non-ACTIVE users cannot log in (`403 ACCOUNT_INACTIVE`).
- **Optional email allowlist** can restrict who may register.

5 · Data Model — Tables by Domain

38 tables. ● in the NN column marks NOT NULL. Types shortened (e.g. `timestampz`). Notes summarise intent and key CHECK constraints.

Identity, Auth & RBAC

users14 cols		
Identity + auth state. Email is citext (case-insensitive unique). Tracks failed_login_count, locked_until, password_changed_at, last_login_at, bloom_opt_in.		
column	type	NN
id	uuid	●
email	citext	●
password_hash	text	●
full_name	text	●
role_id	int2	●
status	text	●
email_verified_at	timestampz	
failed_login_count	int	●
locked_until	timestampz	
password_changed_at	timestampz	●
last_login_at	timestampz	
bloom_opt_in	bool	●
created_at	timestampz	●
updated_at	timestampz	●

roles3 cols		
5 fixed roles. Referenced by users.role_id.		
column	type	NN
id	int2	●
name	text	●
description	text	

permissions3 cols		
43 granular permission codes (e.g. order.approve, invoice.generate).		
column	type	NN
id	int2	●
code	text	●
description	text	

role_permissions2 cols		
Many-to-many role↔permission join — the RBAC matrix.		
column	type	NN
role_id	int2	●
permission_id	int2	●

role_access_windows			6 cols
Optional time-of-day / day-of-week access windows per role (start_time < end_time, day 0–6).			
column	type		NN
id	int		•
role_id	int2		•
day_of_week	int		•
start_time	time		•
end_time	time		•
created_at	timestamptz		•

trade_accounts			15 cols
Business profile for a TRADE_BUYER (1:1 with user). Holds tier, credit_limit, payment_term, account_status, payment_reliability_score (0–5).			
column	type		NN
id	uuid		•
user_id	uuid		•
business_name	text		•
business_reg_no	text		
phone	text		
address	text		
tier_id	int2		•
credit_limit	numeric		•
payment_term	text		•
account_status	text		•
payment_reliability_score	numeric		•
approved_by	uuid		
approved_at	timestamptz		
created_at	timestamptz		•
updated_at	timestamptz		•

auth_sessions			12 cols
Refresh-token sessions (hashed). Supports rotation (rotated_from), revocation, 7-day expiry, max 3 per user.			
column	type		NN
id	uuid		•
user_id	uuid		•
refresh_token_hash	text		•
ip	inet		
user_agent	text		
device_label	text		
issued_at	timestamptz		•
expires_at	timestamptz		•
rotated_from	uuid		
revoked_at	timestamptz		
revoke_reason	text		
created_at	timestamptz		•

email_tokens			7 cols
Single-use tokens for email_verify (24h), password_reset (1h), invitation.			
column	type		NN
id	uuid		•
user_id	uuid		•
purpose	text		•
token_hash	text		•
expires_at	timestamptz		•
used_at	timestamptz		
created_at	timestamptz		•

login_history			8 cols
Append-only audit of every login attempt (success/failure + reason, IP, UA).			
column	type		NN
id	int8		•
user_id	uuid		
email_attempted	citext		•
success	bool		•
failure_reason	text		
ip	inet		
user_agent	text		
occurred_at	timestamptz		•

audit_logs			12 cols
Generic before/after JSONB audit trail keyed by entity_type/entity_id + actor.			
column	type		NN
id	int8		•
actor_id	uuid		
actor_role	text		
action	text		•
entity_type	text		•
entity_id	text		•
before	jsonb		
after	jsonb		
ip	inet		
user_agent	text		
request_id	text		
occurred_at	timestamptz		•

Catalogue & Pricing

categories			6 cols
Self-referential category tree (PRODUCT / SUPPLY / BLOG).			
column	type		NN
id	int		•
name	text		•
parent_id	int		
type	text		•
created_at	timestamptz		•
updated_at	timestamptz		•

suppliers			10 cols
Supplier master with lead_time_days; ACTIVE/INACTIVE.			
column	type		NN
id	int		•
name	text		•
contact_person	text		
phone	text		
email	text		
address	text		
status	text		•
lead_time_days	int		•
created_at	timestamptz		•
updated_at	timestamptz		•

products			17 cols
Catalogue item. stock_qty/reserved_qty/reorder_level, base_price, moq. Type ORCHID/FERTILIZER/SUPPLY/OTHER; status ACTIVE/INACTIVE/DISCONTINUED/OUT_OF_STOCK. CHECK: reserved_qty ≤ stock_qty.			
column	type		NN
id	int		•
sku	text		•
name	text		•
description	text		
category_id	int		•
supplier_id	int		•
product_type	text		•
unit_size	text		
base_price	numeric		•
moq	int		•
stock_qty	int		•
reserved_qty	int		•
reorder_level	int		•
status	text		•
bloom_video_url	text		
created_at	timestamptz		•
updated_at	timestamptz		•

product_images			7 cols
Cloudinary-backed image set with primary flag + sort order.			
column	type		NN
id	int		•
product_id	int		•
cloudinary_public_id	text		
url	text		
is_primary	bool		•
sort_order	int		•
created_at	timestampz		•

bulk_pricing_tiers			5 cols
Quantity-break pricing per product (min_quantity → unit_price).			
column	type		NN
id	int		•
product_id	int		•
min_quantity	int		•
unit_price	numeric		•
created_at	timestampz		•

buyer_tiers			6 cols
SILVER/GOLD/PLATINUM: discount_rate %, credit_cap, priority.			
column	type		NN
id	int2		•
name	text		•
discount_rate	numeric		•
credit_cap	numeric		•
priority	int		•
created_at	timestampz		•

price_history			7 cols
Immutable log of price changes (MANUAL / BULK_IMPORT / PRICE_REQUEST).			
column	type		NN
id	int8		•
product_id	int		•
old_price	numeric		
new_price	numeric		•
changed_by	uuid		
changed_at	timestampz		•
source	text		•

price_change_requests			12 cols
Inventory-raised price changes requiring approval (PENDING/APPROVED/REJECTED).			
column	type		NN
id	int		•
product_id	int		•
requested_by	uuid		•
current_price	numeric		•
requested_price	numeric		•
reason	text		
status	text		•
decided_by	uuid		
decided_at	timestampz		
decision_note	text		
created_at	timestampz		•
updated_at	timestampz		•

Quoting, Cart & Orders

carts			4 cols
One cart per trade account.			
column	type		NN
id	uuid		•
buyer_id	uuid		•
created_at	timestampz		•
updated_at	timestampz		•

cart_items			5 cols
Cart lines (qty > 0).			
column	type		NN
id	int		•
cart_id	uuid		•
product_id	int		•
qty	int		•
created_at	timestampz		•

rfqs			11 cols
Request for Quotation header. DRAFT→SUBMITTED→QUOTED→(CONVERTED EXPIRED DECLINED).			
column	type		NN
id	int		•
rfq_no	text		•
buyer_id	uuid		•
status	text		•
buyer_note	text		
admin_note	text		
quote_expiry	timestampz		
quoted_by	uuid		
quoted_at	timestampz		
created_at	timestampz		•
updated_at	timestampz		•

rfq_items			7 cols
RFQ lines with requested vs quoted unit price.			
column	type		NN
id	int		•
rfq_id	int		•
product_id	int		•
requested_qty	int		•
requested_unit_price	numeric		
quoted_unit_price	numeric		
created_at	timestampz		•

orders			17 cols
Order header. source DIRECT/RFQ_CONVERSION/CART. Status DRAFT→PENDING_APPROVAL→APPROVED→PROCESSING→READY_TO_SHIP→DISPATCHED→DELIVERED (+REJECTED/CANCELLED/RETURNED). Stores subtotal, tier_discount_amount, total.			
column	type	NN	
id	int	●	
order_no	text	●	
buyer_id	uuid	●	
source	text	●	
rfq_id	int		
status	text	●	
subtotal	numeric	●	
tier_discount_amount	numeric	●	
total	numeric	●	
approved_by	uuid		
approved_at	timestampz		
rejection_reason	text		
cancelled_by	uuid		
cancelled_at	timestampz		
cancel_reason	text		
created_at	timestampz	●	
updated_at	timestampz	●	

order_items			8 cols
Order lines, with price_source (BASE/BULK_TIER/RFQ_QUOTE/MANUAL) and frozen unit_price_at_order.			
column	type	NN	
id	int	●	
order_id	int	●	
product_id	int	●	
qty	int	●	
unit_price_at_order	numeric	●	
price_source	text	●	
line_total	numeric	●	
created_at	timestampz	●	

Billing & Returns

invoices			12 cols
Generated from an order. total/paid/balance_due, due_date, status PENDING/PARTIALLY_PAID/PAID/OVERDUE/CANCELLED/VOID.			
column	type		NN
id	int		•
invoice_no	text		•
order_id	int		•
buyer_id	uuid		•
total_amount	numeric		•
paid_amount	numeric		•
balance_due	numeric		•
due_date	date		•
status	text		•
pdf_url	text		
created_at	timestampz		•
updated_at	timestampz		•

invoice_adjustments			7 cols
Credit/debit adjustments to an invoice, optionally linked to an RMA.			
column	type		NN
id	int		•
invoice_id	int		•
amount	numeric		•
reason	text		•
rma_id	int		
created_by	uuid		•
created_at	timestampz		•

payments			12 cols
Payments against invoices. method BANK_TRANSFER/CHEQUE/CASH/ONLINE/CREDIT_NOTE; reversible.			
column	type		NN
id	int		•
payment_no	text		•
invoice_id	int		•
buyer_id	uuid		•
amount	numeric		•
method	text		•
reference	text		
recorded_by	uuid		•
received_at	timestampz		•
reversed_at	timestampz		
reversal_reason	text		
created_at	timestampz		•

rma_requests			15 cols
Return / refund request. reason_category enum; status PENDING/APPROVED/REJECTED/RESOLVED/CANCELLED; evidence_urls[].			
column	type		NN
id	int		•
rma_no	text		•
order_id	int		•
buyer_id	uuid		•
status	text		•
reason_category	text		•
reason_detail	text		
evidence_urls	text[]		
decided_by	uuid		
decided_at	timestampz		
resolution	text		
inventory_adjustment_note	text		
invoice_adjustment_note	text		
created_at	timestampz		•
updated_at	timestampz		•

rma_items			5 cols
RMA lines referencing original order_items.			
column	type		NN
id	int		•
rma_id	int		•
order_item_id	int		•
qty	int		•
created_at	timestampz		•

Fulfilment & Inventory

deliveries			11 cols
One delivery per order. status PENDING→ASSIGNED→DISPATCHED→IN_TRANSIT→DELIVERED (+FAILED/CANCELLED). pod_url + buyer_confirmed_at.			
column	type		NN
id	int		•
order_id	int		•
assigned_to	uuid		
status	text		•
dispatch_date	timestampz		
pod_url	text		
pod_uploaded_at	timestampz		
buyer_confirmed_at	timestampz		
failure_note	text		
created_at	timestampz		•
updated_at	timestampz		•

delivery_events			6 cols
Per-delivery status timeline with actor + note.			
column	type		NN
id	int		•
delivery_id	int		•
status	text		•
note	text		
actor_id	uuid		
occurred_at	timestampz		•

stock_movements			9 cols
Immutable inventory ledger (INITIAL/PURCHASE/ORDER_RESERVE/ORDER_RELEASE/ORDER_FULFILL/RMA_RETURN/MANUAL_ADJUSTMENT/DAMAGE_WRITE_OFF/STOCKTAKE)			
column	type		NN
id	int8		•
product_id	int		•
movement_type	text		•
qty	int		•
ref_table	text		
ref_id	text		
performed_by	uuid		
note	text		
occurred_at	timestampz		•

stock_alerts			9 cols
LOW_STOCK/OUT_OF_STOCK/OVERSTOCK/REORDER alerts; OPEN→ACKNOWLEDGED→RESOLVED.			
column	type		NN
id	int		•
product_id	int		•
alert_type	text		•
threshold_value	int		
generated_at	timestampz		•
status	text		•
acknowledged_by	uuid		
acknowledged_at	timestampz		
created_at	timestampz		•

Content, Engagement & Ops

cms_blocks			8 cols
Editable site content blocks (HERO/BANNER/TEXT/HTML/MARKDOWN/JSON), publish flag.			
column	type		NN
id	int		•
key	text		•
type	text		•
content	jsonb		•
is_published	bool		•
updated_by	uuid		
created_at	timestampz		•
updated_at	timestampz		•

bloom_events			6 cols
"Bloom" engagement analytics — product smile_score events (opt-in).			
column	type		NN
id	int8		•
user_id	uuid		
product_id	int		•
session_anon_id	text		
smile_score	numeric		
triggered_at	timestampz		•

notifications_outbox			11 cols
Transactional email outbox with retry (attempts, next_attempt_at, last_error).			
column	type		NN
id	int8		•
recipient_email	text		
recipient_user_id	uuid		
template	text		•
payload	jsonb		•
status	text		•
attempts	int		•
next_attempt_at	timestampz		
last_error	text		
sent_at	timestampz		
created_at	timestampz		•

job_runs			6 cols
Execution log for scheduled jobs (outcome + meta).			
column	type		NN
id	int8		•
job_name	text		•
started_at	timestampz		•
finished_at	timestampz		
outcome	text		
meta	jsonb		

settings			3 cols
Key/value (JSONB) platform settings — businessName, lowStockThreshold, sessionCap.			
column	type		NN
key	text		•
value	jsonb		•
updated_at	timestampz		•

6 · Status Vocabularies (CHECK constraints)

Status / enum-like columns are enforced by DB CHECK constraints — the canonical source of valid values:

Column	Allowed values
users.status	ACTIVE · LOCKED · SUSPENDED · INACTIVE
trade_accounts.account_status	PENDING_APPROVAL · ACTIVE · SUSPENDED · CLOSED
trade_accounts.payment_term	NET_15 · NET_30 · NET_45 · NET_60
products.status	ACTIVE · INACTIVE · DISCONTINUED · OUT_OF_STOCK
products.product_type	ORCHID · FERTILIZER · SUPPLY · OTHER
rfqs.status	DRAFT · SUBMITTED · QUOTED · EXPIRED · DECLINED · CONVERTED
orders.status	DRAFT · PENDING_APPROVAL · APPROVED · REJECTED · PROCESSING · READY_TO_SHIP · DISPATCHED · DELIVERED · CANCELLED · RETURNED
orders.source	DIRECT · RFQ_CONVERSION · CART
order_items.price_source	BASE · BULK_TIER · RFQ_QUOTE · MANUAL
invoices.status	PENDING · PARTIALLY_PAID · PAID · OVERDUE · CANCELLED · VOID
payments.method	BANK_TRANSFER · CHEQUE · CASH · ONLINE · CREDIT_NOTE
rma_requests.status	PENDING · APPROVED · REJECTED · RESOLVED · CANCELLED
rma_requests.reason_category	DAMAGED · WRONG_ITEM · QUALITY_ISSUE · SHORT_SHIPPED · LATE_DELIVERY · BUYER_REMORSE · OTHER
deliveries.status	PENDING · ASSIGNED · DISPATCHED · IN_TRANSIT · DELIVERED · FAILED · CANCELLED
stock_movements.movement_type	INITIAL · PURCHASE · ORDER_RESERVE · ORDER_RELEASE · ORDER_FULFILL · RMA_RETURN · MANUAL_ADJUSTMENT · DAMAGE_WRITE_OFF · STOCKTAKE_CORRECTION
stock_alerts (type/status)	LOW_STOCK · OUT_OF_STOCK · OVERSTOCK · REORDER / OPEN · ACKNOWLEDGED · RESOLVED
price_change_requests.status	PENDING · APPROVED · REJECTED

7 · Core Business Workflows

RFQ → Order → Invoice → Payment

RFQ DRAFT→SUBMITTED → admin QUOTED (quote_expiry) → buyer accepts → CONVERTED to order
(source=RFQ_CONVERSION) → order PENDING_APPROVAL → APPROVED → PROCESSING → READY_TO_SHIP → DISPATCHED →
DELIVERED → invoice generated (NET terms) → PARTIALLY_PAID/PAID

Pricing precedence on an order line (`order_items.price_source`): RFQ_QUOTE → BULK_TIER (quantity break) → BASE, with the chosen unit price frozen as `unit_price_at_order` . Buyer-tier discount (SILVER 3% / GOLD 5% / PLATINUM 7%) is captured in `orders.tier_discount_amount` .

Inventory reservation ledger

order placed → ORDER_RESERVE (reserved_qty↑) → fulfil → ORDER_FULFILL (stock_qty↓) → cancel → ORDER_RELEASE
(reserved_qty↓) → return → RMA_RETURN (stock_qty↑)

Every change is an immutable `stock_movements` row; `products.reserved_qty ≤ stock_qty` is enforced by CHECK. Low stock vs `reorder_level` raises `stock_alerts` .

Delivery

PENDING → ASSIGNED (coordinator) → DISPATCHED → IN_TRANSIT → DELIVERED (pod_url upload) → buyer confirms
(buyer_confirmed_at) | FAILED (failure_note)

Returns (RMA) & pricing approval

RMA PENDING → finance/admin APPROVED/REJECTED → RESOLVED → optional invoice_adjustment (credit note) + stock
RMA_RETURN

Inventory raises price_change_request PENDING → admin/inventory price.approve → APPROVED →
products.base_price updated + price_history row

8 · Scheduled Jobs

node-cron jobs (Asia/Colombo TZ), gated by `ENABLE_CRON` ; runs logged to `job_runs` :

Job	Schedule	Does
outboxDispatch	every 30s	Sends pending <code>notifications_outbox</code> emails with retry/backoff.
stockCheck	daily 07:00	Scans stock vs reorder level, raises/updates <code>stock_alerts</code> .
invoiceAging	daily 08:00	Flags overdue invoices, queues payment reminders.
quoteExpiry	daily 08:00	Expires RFQs past <code>quote_expiry</code> (QUOTED→EXPIRED).
sessionSweep	daily 02:00	Purges expired/revoked <code>auth_sessions</code> .

9 · Edge Cases & Known Inconsistencies

State-machine vs DB status drift. `src/utils/stateMachine.js` defines transition tables using a *different* status vocabulary and role names than the DB CHECK constraints — e.g. it uses order `PENDING/CONFIRMED` and roles like `ORDER_MANAGER/SALES_MANAGER/LOGISTICS_COORDINATOR`, whereas the DB uses `PENDING_APPROVAL/APPROVED/...` and roles `ADMIN/INVENTORY_MANAGER/FINANCE_OFFICER/DELIVERY_COORDINATOR`. Reconcile before relying on `assertTransition`.

- **Coarse middleware.** Next.js middleware only checks for a refresh-cookie's presence; true role/permission enforcement is server-side in the API. A user manually visiting another portal's URL gets data only if the API permits it — UI may render an empty/locked shell.
- **Inactive accounts retained.** Several seeded `INVENTORY_MANAGER` users are `INACTIVE` (auto/smoke-test accounts) and cannot log in — expected, but they inflate the user table.
- **Citext email.** `users.email` and `login_history.email_attempted` are `citext` — logins are case-insensitive; uniqueness is enforced case-insensitively.
- **Reserved-stock invariant.** Orders cannot reserve beyond `stock_qty` (CHECK `reserved_qty ≤ stock_qty`); concurrent orders must serialize reservation to avoid oversell.
- **Invoice positivity.** `invoices.total_amount > 0` CHECK means a fully-credited (zero) invoice must use status `VOID/CANCELLED` rather than a 0 total.
- **Payment reversal.** Payments are not deleted; reversal sets `reversed_at / reversal_reason`, so invoice balances must recompute from non-reversed payments.
- **Refresh-token reuse.** A refresh token not found in `auth_sessions` is rejected (`401 INVALID_REFRESH_TOKEN`) — basis for reuse-detection hardening.
- **Email delivery is best-effort.** Registration/reset proceed even if SMTP send fails (errors are logged, not surfaced); real delivery depends on the outbox worker + mailer config.

10 · Appendix — Settings & Tiers

Platform settings (settings)

key	value
businessName	"K ORCHIDS"
lowStockThreshold	10
sessionCap	3

Buyer tiers (buyer_tiers)

Tier	Discount	Credit cap	Priority
SILVER	3%	500,000	1
GOLD	5%	1,000,000	2
PLATINUM	7%	2,500,000	3

Migrations: 0001_extensions · 0002_roles_permissions · 0003_users_auth · 0004_trade_catalogue · 0005_pricing_rfq_cart_orders · 0006_invoices_payments_rma_delivery · 0007_crosscutting · 0008_indexes_constraints.